

# GaN Doherty Amplifiers for Backhaul Radio Links

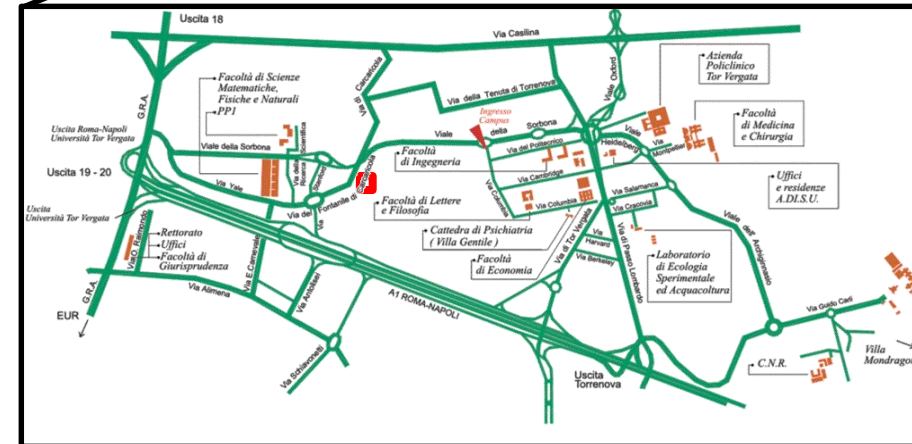
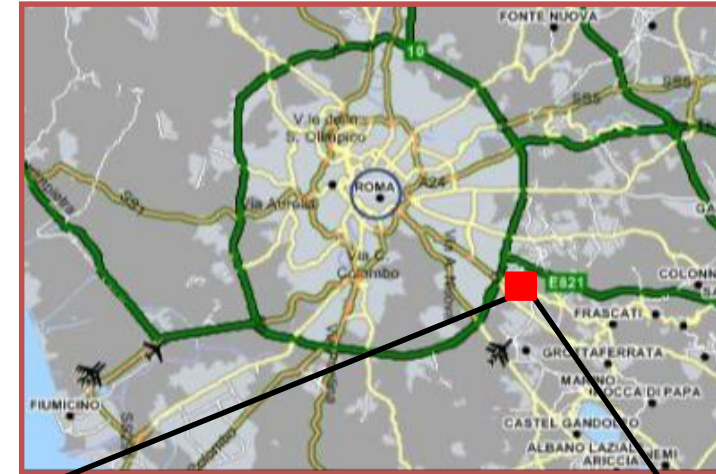
**Rocco Giofrè**

*E.E.Dept., University of Rome “Tor Vergata”, Rome, Italy*

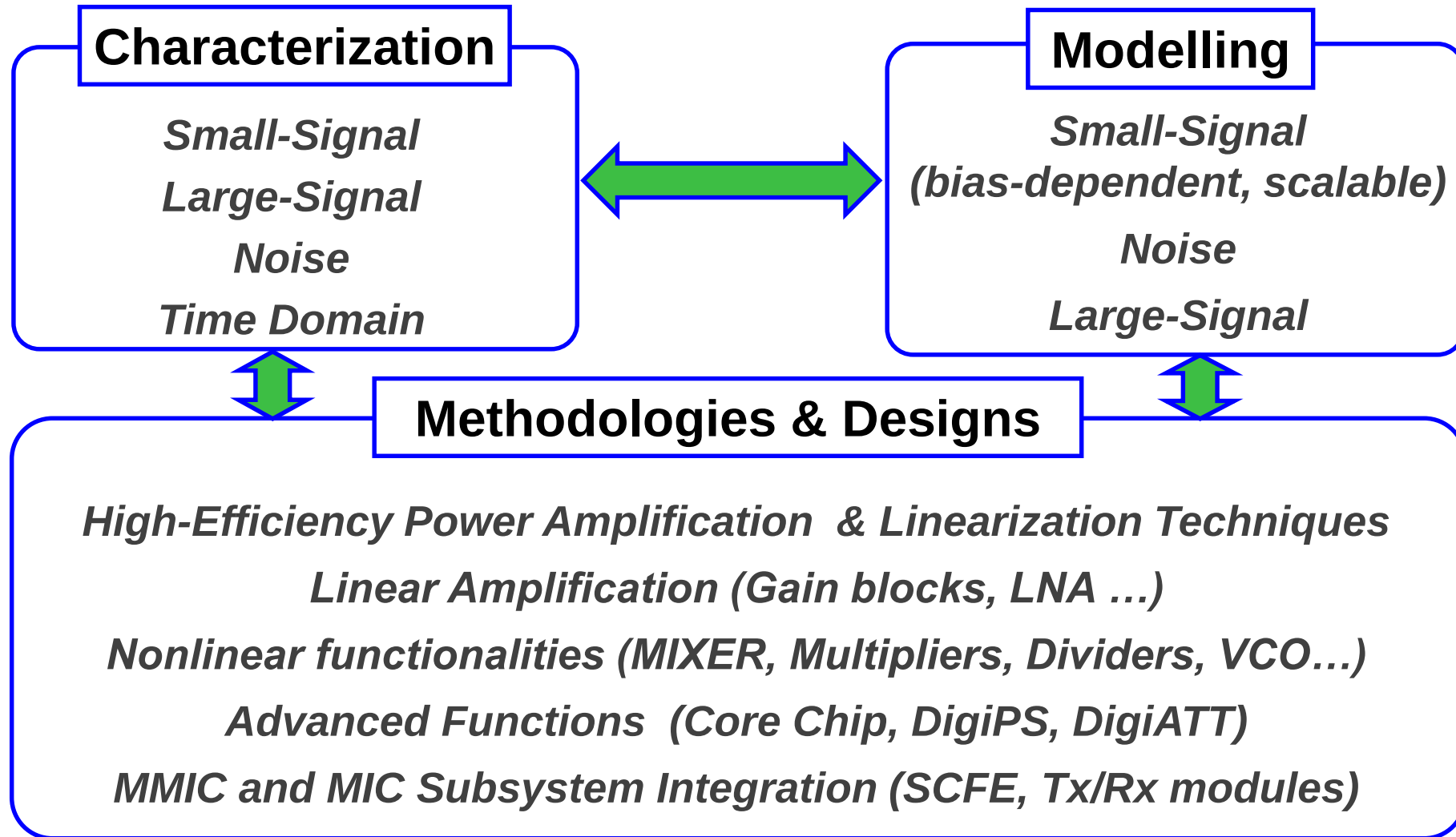
[giofr@ing.uniroma2.it](mailto:giofr@ing.uniroma2.it)

# Who and Where we are

- **University of Roma Tor Vergata**
  - Founded in 1982
  - 600 hectares of Campus
  - 6 Faculties & 50k students
- **Electronic Engineering Department**
  - Founded in 1983
  - 150 people
- **MiMEG Group**
  - 1 Full & 3 Associate Prof.
  - 5 Researchers
  - 2 Technicians
  - 8 PhD Students
  - 5 Post Doc



# What we do at MiMEG

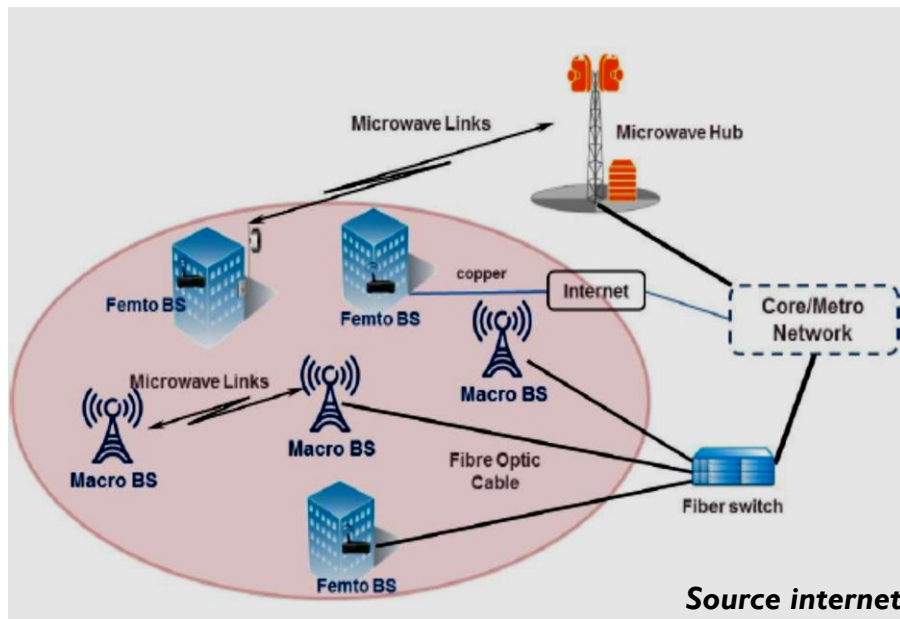


***From 250kHz up to 150 GHz and in many technologies***

- Backhaul Radio Links (BRL)
- Challenges in 5G
- Why Doherty (DPA)?
- Why Gallium Nitride?
- DPA Issues & Solutions
- **Focus on DPA Linearity Issues**
  - *Proposed Approach*
  - *MMIC Design*
  - *Experimental Results*
- Conclusion

# Backhaul Radio Links (BRL)

- Backhaul Links are the connections between the backbone networks and the peripheral subnetworks.
  - *Over 50% of the world's mobile base stations are connected using **point-to-point microwave technologies from 6GHz to 40GHz and beyond.***

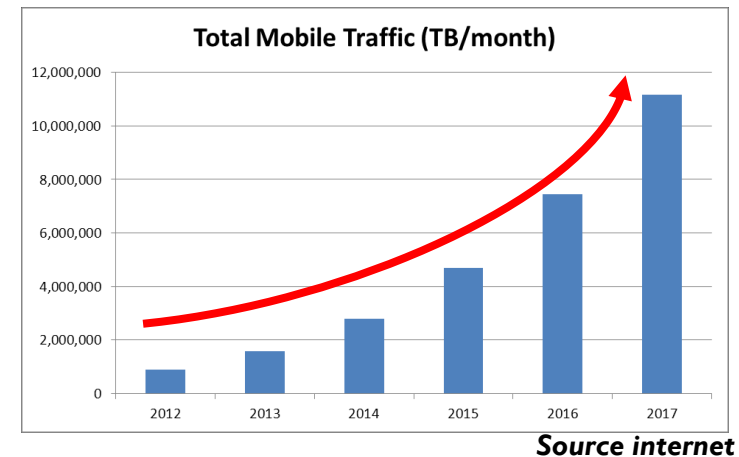
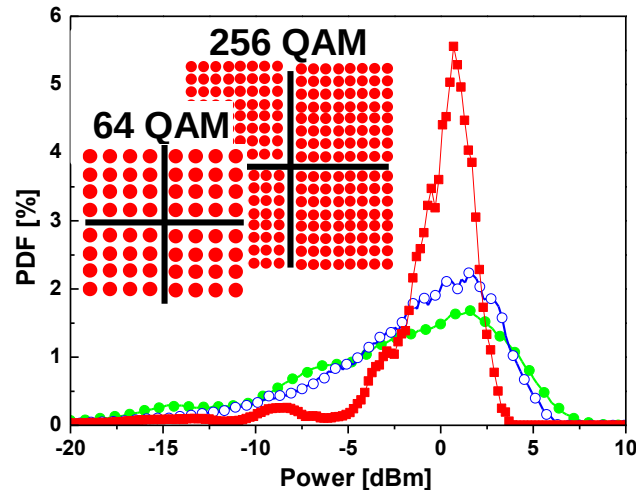


## Why Backhaul Radio Links ?

- ☺ *Inherently more secure than cable networks.*
- ☺ *Rapidly rolling out*
- ☺ *Cost-effective.*

- **5 million microwave links** in operation.
- **\$10 Billion per year** for the next mobile infrastructures.

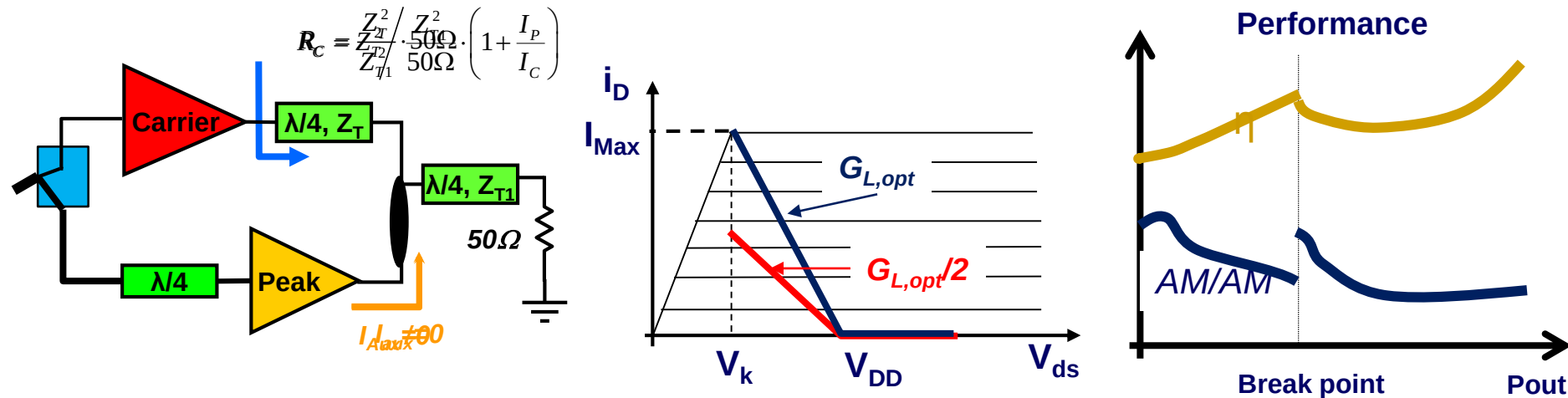
- **5G Promises:** All on Clouds, Ubiquitous Connectivity & Smart Infrastructures
- **5G Expectations:** latency<1ms, Gbps data rates everywhere, 100x network energy efficiency and >10 Mbps/m<sup>2</sup> capacity.
- **Electronics key Drivers:** Massive MIMO, larger bandwidth, higher efficiency and frequency of operation and better linearity.



- In this scenario, Power Amplifiers with **high efficiency & linearity** over a **wide frequency range and output power** are essential in every node of the network.

# Why Doherty?

- The DPA is becoming the ***solution of choice*** in BRL and elsewhere, thanks to its ***efficiency vs. output power*** characteristic.

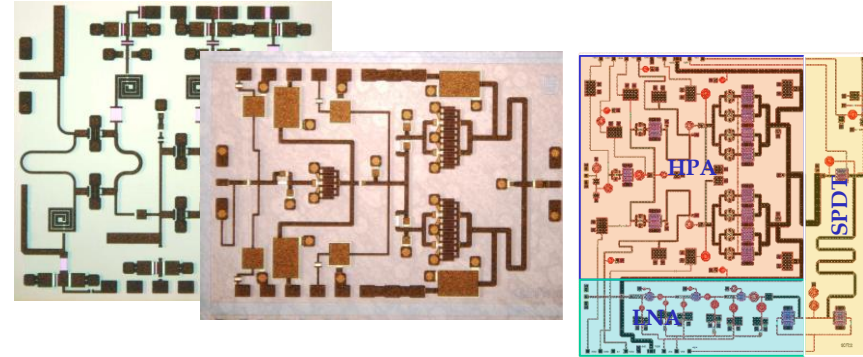


- This feature was the triggering cause of research activities aiming to improve DPA performance in terms of gain, bandwidth, linearity and also efficiency.

# Why Gallium Nitride?

- **GaN offers the possibility to realize many MMIC functionalities:**

- Low Noise Amplifiers
- High Power Amplifiers
- Switches
- Single Chip Front-End
- Also digital circuits such as SIPO



- **GaN is emerging as key technology also for high frequency PAs**

- High power density,
- High Power Added Efficiency (PAE)
- Better thermal properties
- Good Gain
- Easier impedance matching

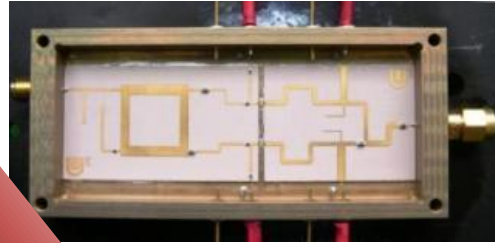
*Many foundries  
offer GaN technology!*

The basic DPA architecture is affected by several issues, which have been investigated and addressed by suitable design solutions

- **Low efficiency @break with respect to saturation**
  - 👍 *Harmonic tuning strategies*
  - 👍 *Accounting for device  $R_{ON}$*
- **Low Gain due to the high input power splitter factor**
  - 👍 *Different drain bias voltages*
  - 👍 *Asymmetrical driver stage*
- **Limited bandwidth due to  $\lambda/4$** 
  - 👍 *Development of new wide-band output combining architectures (BW=15%, 36%, 83%)*
  - 👍 *Development of multi-band architectures (dual-, tri-band)*
- **Limited efficiency for back-off >6dB**
  - 👍 *Multi-step, multi-way approaches*
- **Higher distortion with respect to class AB PA**
  - 👍 *Adoption of innovative design techniques (mismatch, nonlinear driver)*

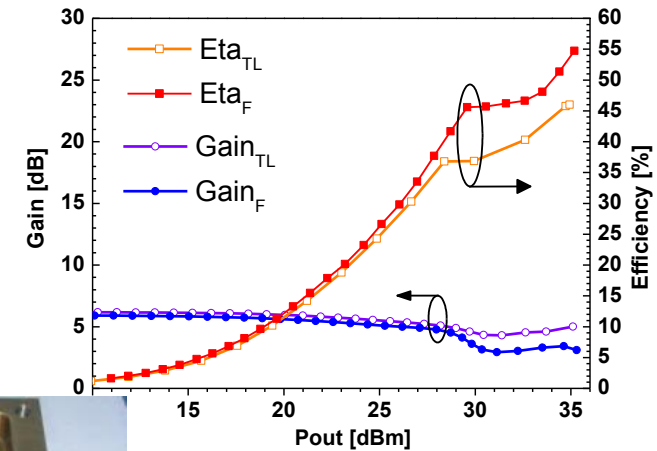
# DPA Efficiency Improvements

**GaN**

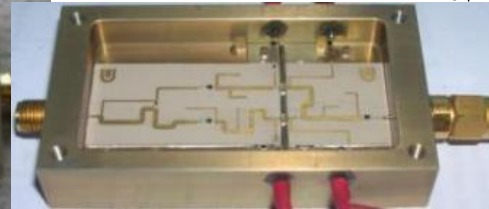


**Performances**

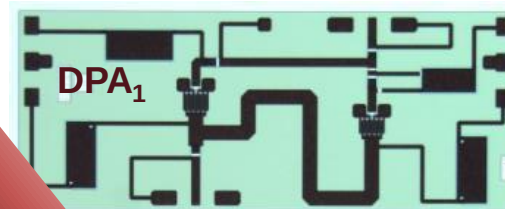
- $f_1 = 2.14 \text{ GHz}$
- $P_{out} > 4 \text{ W}$
- $\eta > 50\%$  (6dB OBO)



**Class F**



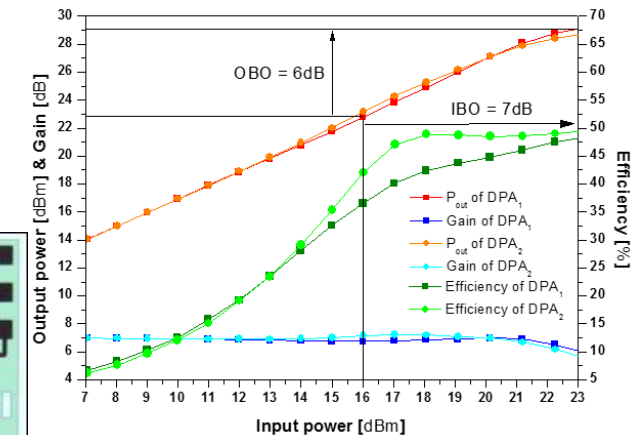
**GaN**



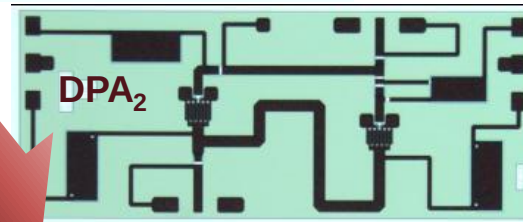
**GaAs**

**Performances**

- $f_1 = 9.6 \text{ GHz}$
- $P_{out} = 29 \text{ dBm}$
- $\eta > 55\%$  (6dB OBO)

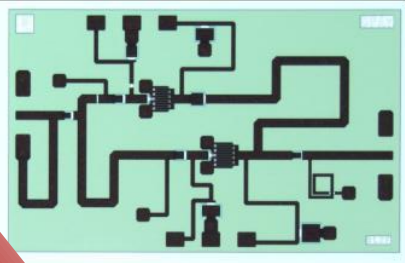


**Acc. Ron**



# DPA Gain Improvements

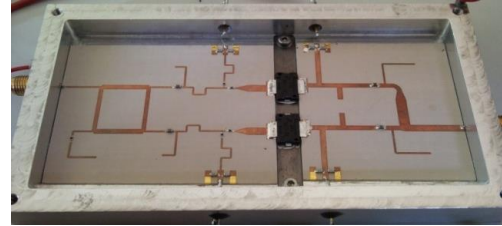
GaAs



Performances

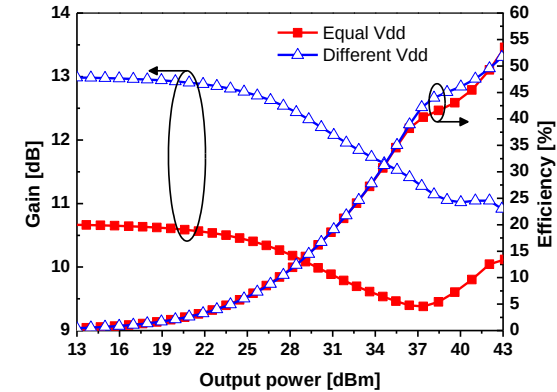
- $f = 2.14$  GHz
- $P_{out} > 4W$
- $\eta > 50\%$  (6dB OBO)

LDMOS



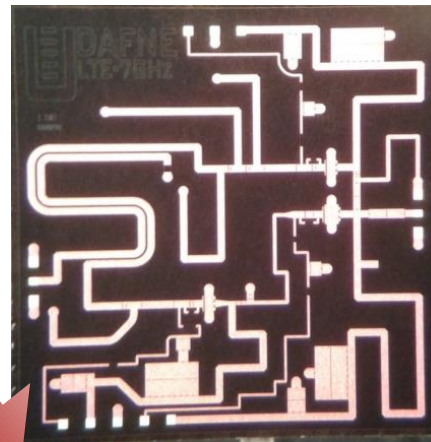
Performances

- $f = 2.14$  GHz
- $P_{sat} > 20$  W
- $\eta > 40\%$  (6dB OBO)

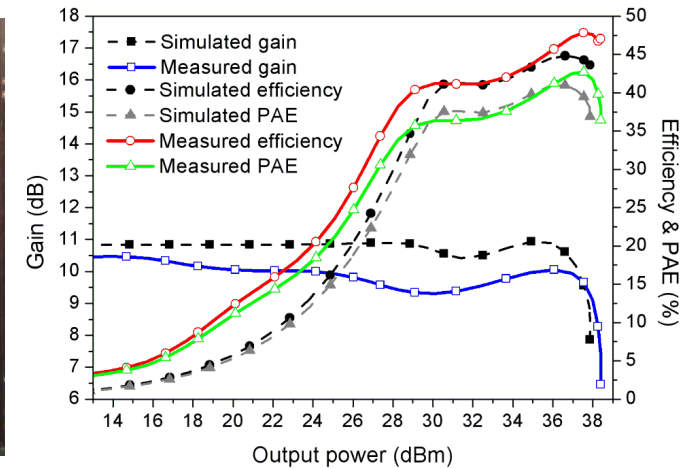


Performances

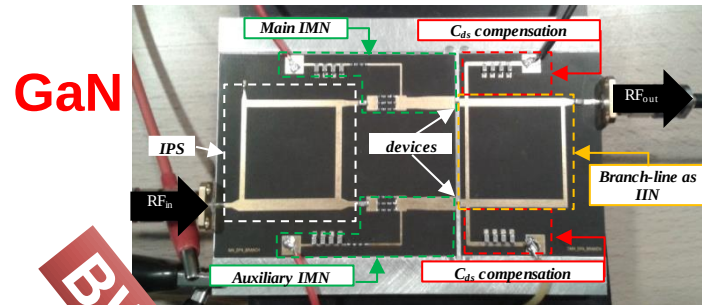
- $f = 7$  GHz
- $P_{sat} > 6$  W
- $\eta > 40\%$  (7dB OBO)
- Gain  $> 10$  dB



GaN

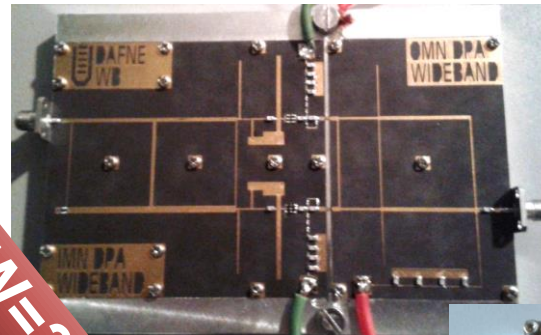


# DPA Bandwidth Improvement



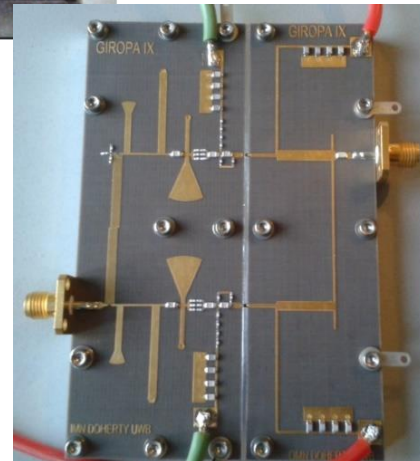
## Performances

- $f = 1.95 - 2.25$  GHz (15%)
- $P_{out} > 15$  W
- $\eta > 65\%$



## Performances

- $f = 1.67 - 2.41$  GHz (36%)
- $P_{out} > 15$  W
- $\eta > 53\%$

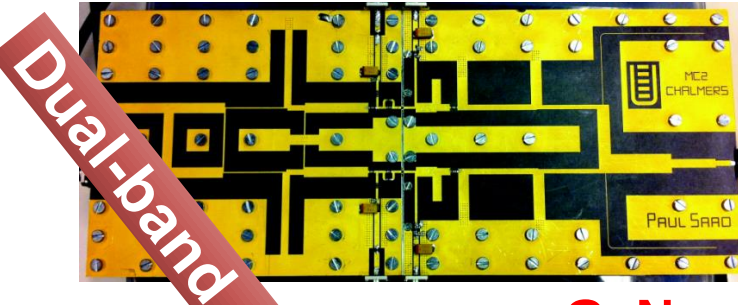


## Performances

- $f = 1.05 - 2.55$  GHz (83%)
- $P_{out} > 10$  W
- $\eta > 40\%$  (6dB OBO)

**GaN**

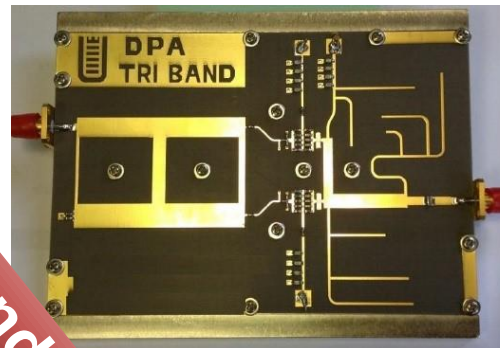
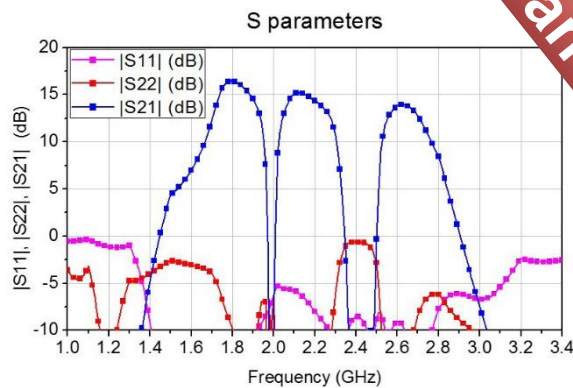
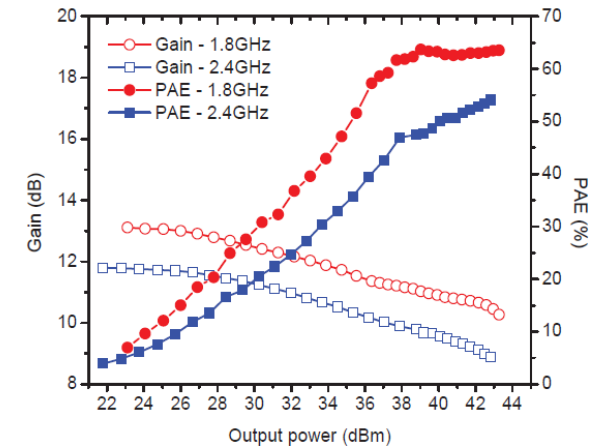
# DPA multiband solutions



GaN

## Performances

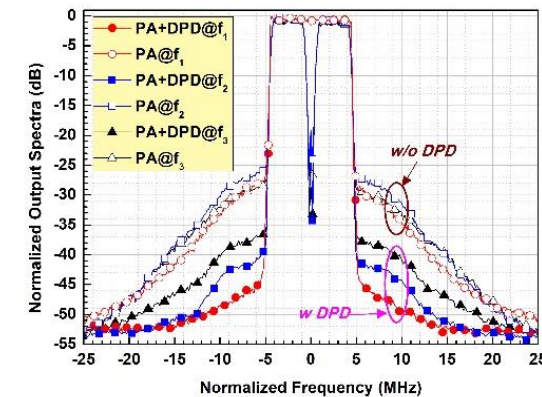
- Pout = 20 W
- Freq: 1,8GHz - 2,4 GHz
- PAE: 65% - 50%



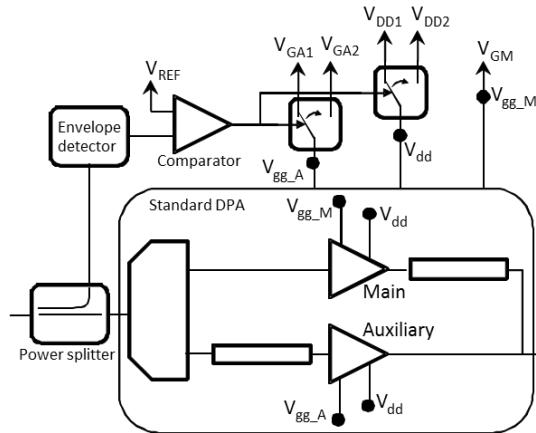
GaN

## Performances

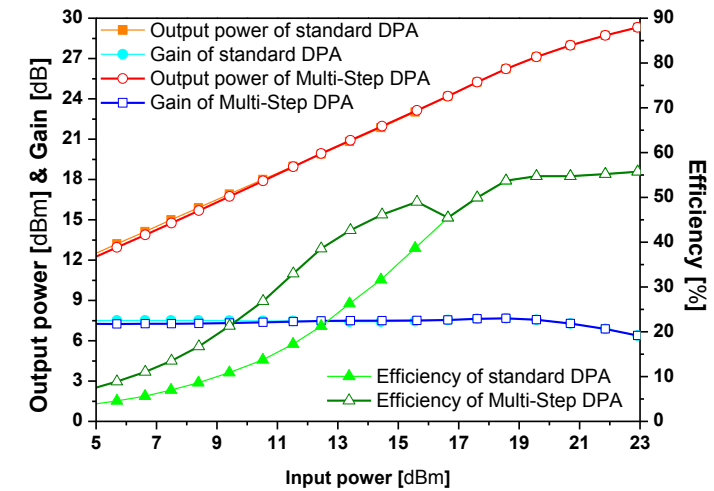
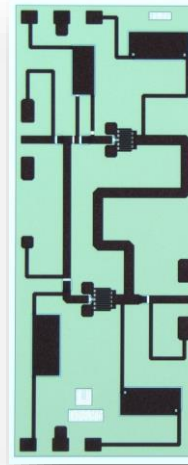
- Freq:  $f_1=1,84\text{GHz}$  –  $f_2=2,14\text{ GHz}$  –  $f_3=2,65\text{GHz}$
- 10MHz-LTE @  $f_1$
- 10MHz-WCDMA @  $f_2$  & @  $f_3$



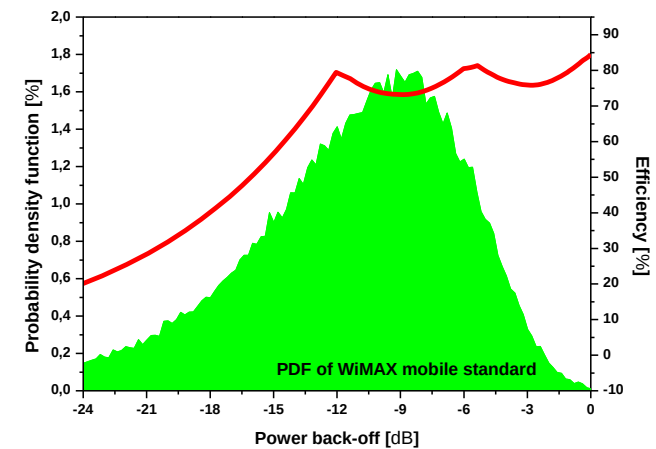
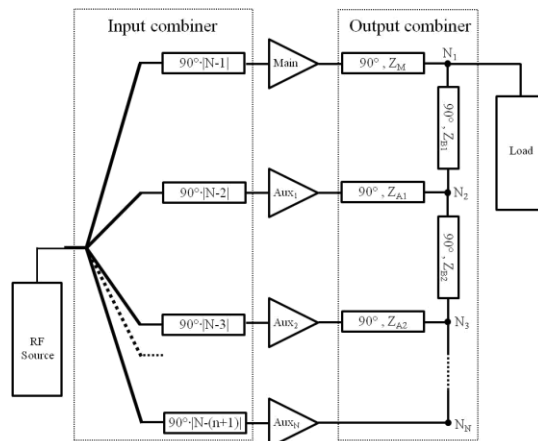
## Multi-step approach



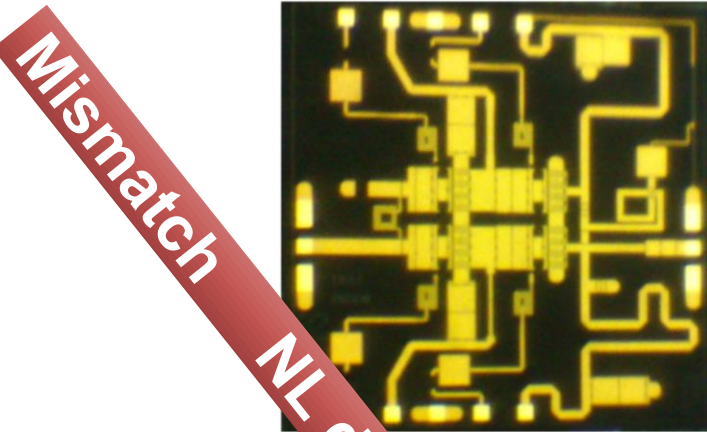
GaAs



## Multi-Stage approach



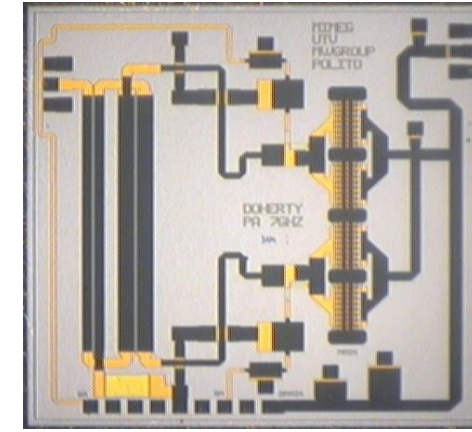
# DPA Linearity Improvements



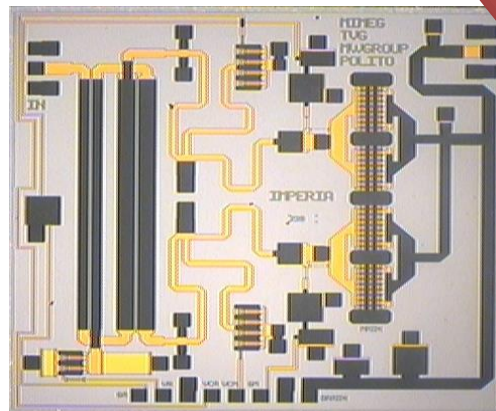
GaN

## Performances

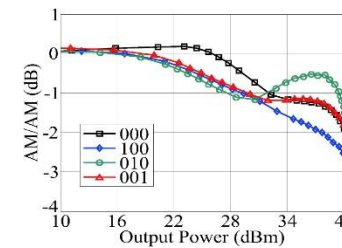
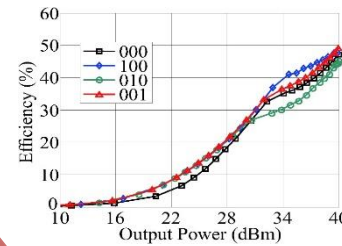
- $f = 7 \text{ GHz}$
- $P_{\text{sat}} > 6 \text{ W}$
- $\eta > 40\%$  (7dB OBO)
- Gain = 10 dB
- AM/PM < 3°



GaN

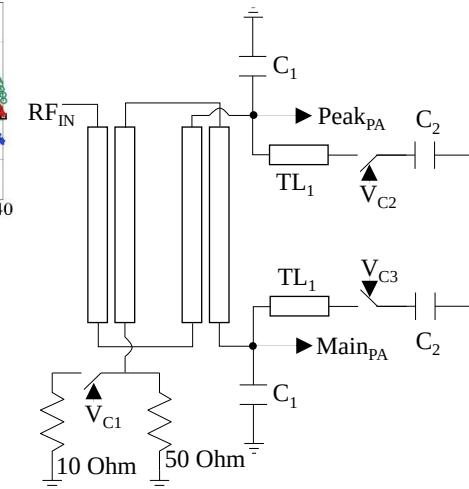


GaN



## Performances

- $f = 6.65\text{-}7.35 \text{ GHz}$  (BW=10%)
- $P_{\text{sat}} > 10 \text{ W}$
- $\eta > 30\text{-}40\%$  (7dB OBO)



- Studies on the distortion generation mechanisms in GaN devices have shown that:
  - The nonlinear behavior of  $g_m$  is the main responsible of the **AM/AM**.
  - The nonlinear variation of  $Z_{in}$  is the main responsible of the **AM/PM**.
  - The **AM/PM** tends to be opposite to the **AM/AM**

## Input Capacitance

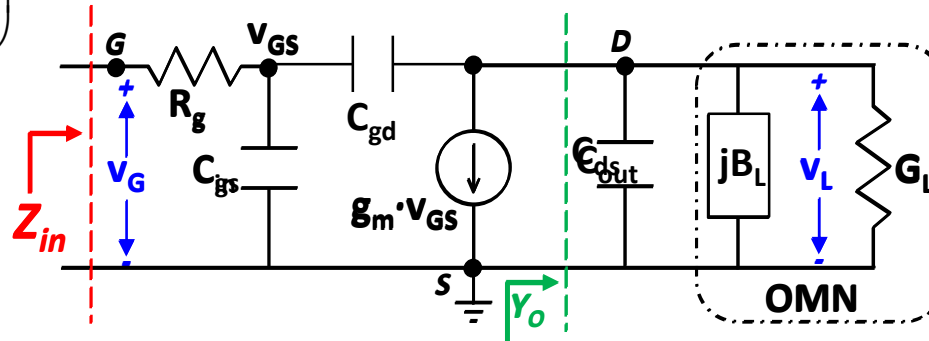
$$C_{in} = C_{gs} + C_{gd} \left( 1 + \frac{g_m}{G_L} \right)$$

## Output Capacitance

$$C_{out} = C_{ds} + C_{gd} \left( 1 + \frac{G_L}{g_m} \right)$$

## Input Impedance

$$Z_{in} = R_G - \frac{j}{\omega \cdot C_{in}}$$



HEMT Simplified model  
After Miller's Theorem

## Output Admittance

$$Y_o = G_L + j(B_L + \omega \cdot C_{out})$$

- Under this assumptions, the voltage amplification is given by:

$$v_L = -\frac{g_m}{Y_o \cdot j\omega C_{in} \cdot Z_{in}} \cdot v_G$$

- The AM/PM distortion, i.e., the phase variation of  $v_L$  with respect to the  $v_G$  amplitude, can be derived:

$$AM/PM = -\Delta\angle Z_{in} - \Delta\angle Y_o \simeq -\Delta\angle Z_{in}$$

- Indeed, both  $g_m$  and  $j\omega C_{in}$  introduce a constant contribution and  $\Delta\angle Y_o \simeq 0$

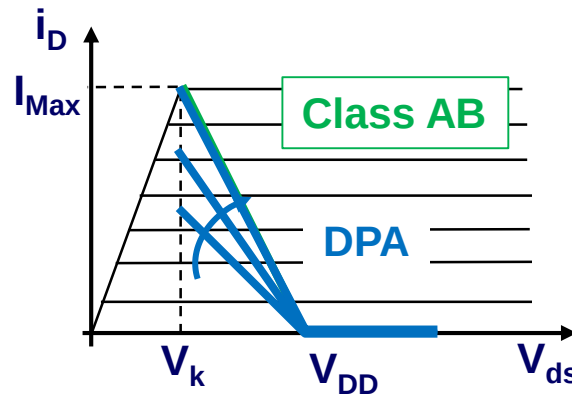
$$Y_o = G_L + j(B_L + \omega \cdot C_{out}) = G_L + j\left[B_L + \omega \cdot (C_{ds} + C_{gd} + C_{gd} \cdot G_L / g_m)\right] \simeq G_L$$

- $C_{ds}$  and  $C_{gd}$  show negligible non-linearities all along the RF input signal.
- The ratio  $G_L / g_m \ll 1$  even while, in the “Doherty region”,  $G_L$  is modulated.

$$C_{out} = C_{ds} + C_{gd} + \cancel{C_{gd} \cdot G_L / g_m} = C_{ds} + C_{gd} \quad \Rightarrow \quad B_L = -\omega \cdot C_{out} \quad \Rightarrow \quad \text{Im}(Y_o) = 0$$

- Therefore, the variation of  $Z_{in}$  represents the dominant contribution into the overall AM/PM.

$$AM/PM = -\Delta\angle Z_{in} - \Delta\angle Y_o \simeq -\Delta\angle Z_{in} = \Delta\angle \left( \frac{1}{R_G \cdot (C_{gs} + C_{gd} + C_{gd} \cdot g_m / G_L)} \right)$$



Output Conductance  
 dependency  
 $g_m/G_L \gg 1$

$G_L = \text{cost}$   $\Rightarrow$  Limited variation of  $Z_{in}$   $\Rightarrow$

AM/PM is a function of the device nonlinearities ( $C_{gs}$ ,  $C_{ds}$ ,  $C_{gd}$ ,  $g_m$ ) only.

$G_L \neq \text{cost}$   $\Rightarrow$  Larger variation of  $Z_{in}$   $\Rightarrow$

AM/PM is a function of  $C_{gs}$ ,  $C_{ds}$ ,  $C_{gd}$ ,  $g_m$  and  $G_L$ , due to the presence of the term  $C_{gd}g_m/G_L$

# Proposed Approach

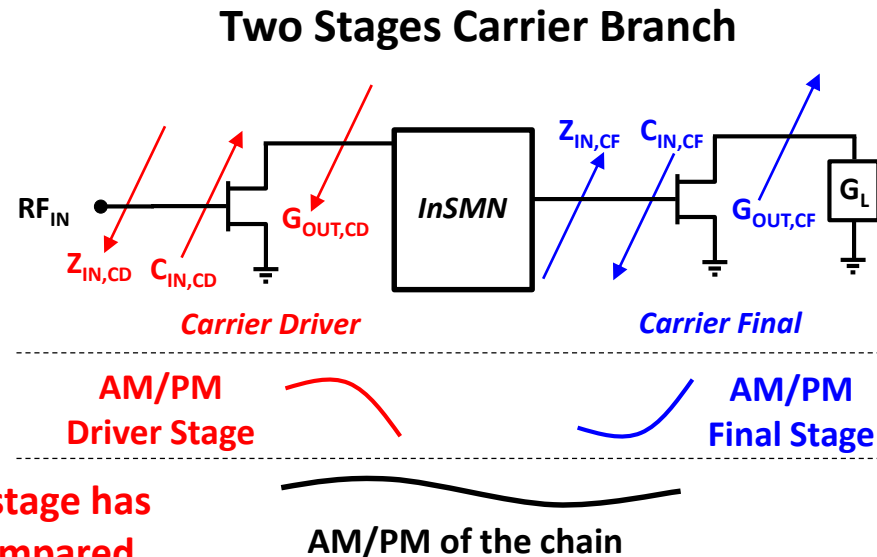
- On these bases, the AM/PM of a GaN DPA can be minimized directly at chip level.
- Focusing on the Carrier branch, that has been demonstrated to be the dominant contributor onto the AM/PM of the overall DPA.

$$G_{L,CD} \downarrow \text{ since } Z_{in,CF} \uparrow$$

$$C_{in} = C_{gs} + C_{gd} \left( 1 + \frac{g_m}{G_{out}} \right)$$

$$Z_{in} = R_G - \frac{j}{\omega \cdot C_{in}}$$

**$Z_{in}$  & AM/PM of the Driver stage has an opposite behavior as compared to the ones of the Final Stage**



$$G_{out} = \frac{G_{L,opt}}{2} \rightarrow G_{L,opt}$$

$$C_{in} = C_{gs} + C_{gd} \left( 1 + \frac{g_m}{G_{out}} \right)$$

$$Z_{in} = R_G - \frac{j}{\omega \cdot C_{in}}$$

**Final stage undergoes a voltage gain compression while  $G_L$  is modulated**

- Clearly, the InSMN between final and driver stage represents the key point of this approach. **It has to be minimal and without an impedance inverting behavior.**

- The design strategy described above has been adopted to realize a GaN MMIC DPA oriented to microwave backhaul links applications.
- The two stages approach is used in both Carrier and Peaking branches.

## DPA Requirements

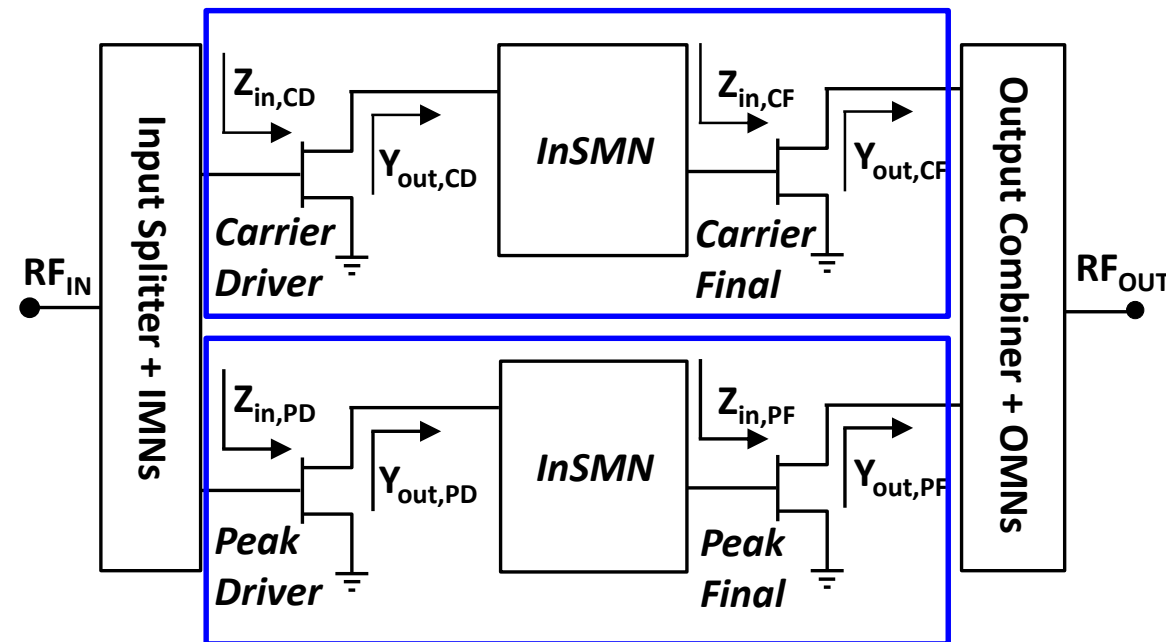
- $OBO = 7\text{dB}$
- $\text{Freq} = 7\text{GHz}$
- $P_{\text{sat}} > 37\text{dBm}$

## Design Steps

- Active chain design
- Output Combiner
- Input Splitter

## MMIC Technology

- $0.25\mu\text{m}$  GaN Qorvo



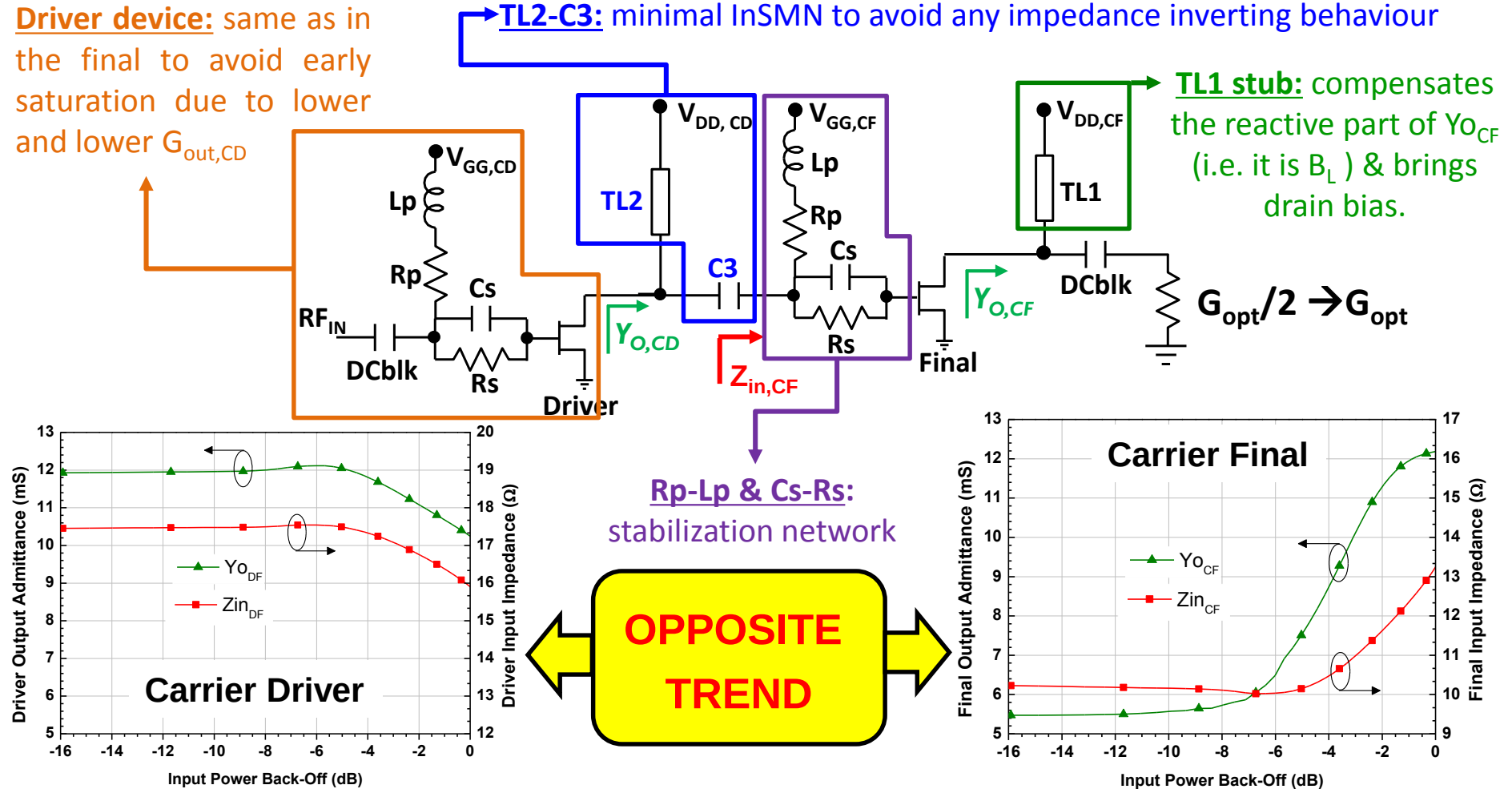
**DPA Topology**

- Final stage: 8x100  $\mu\text{m}$  HEMT to achieve a DPA  $P_{\text{sat}} > 37$  dBm.

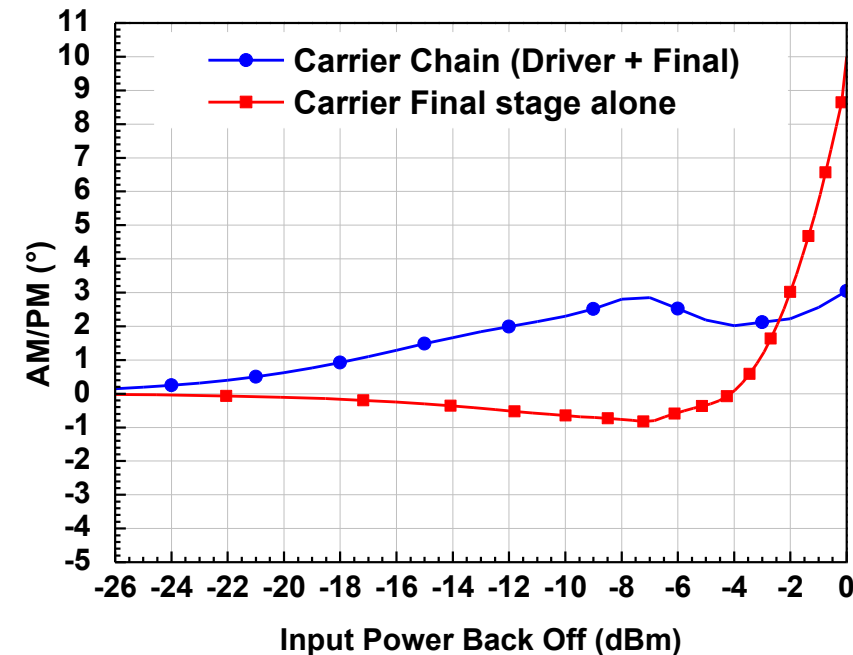
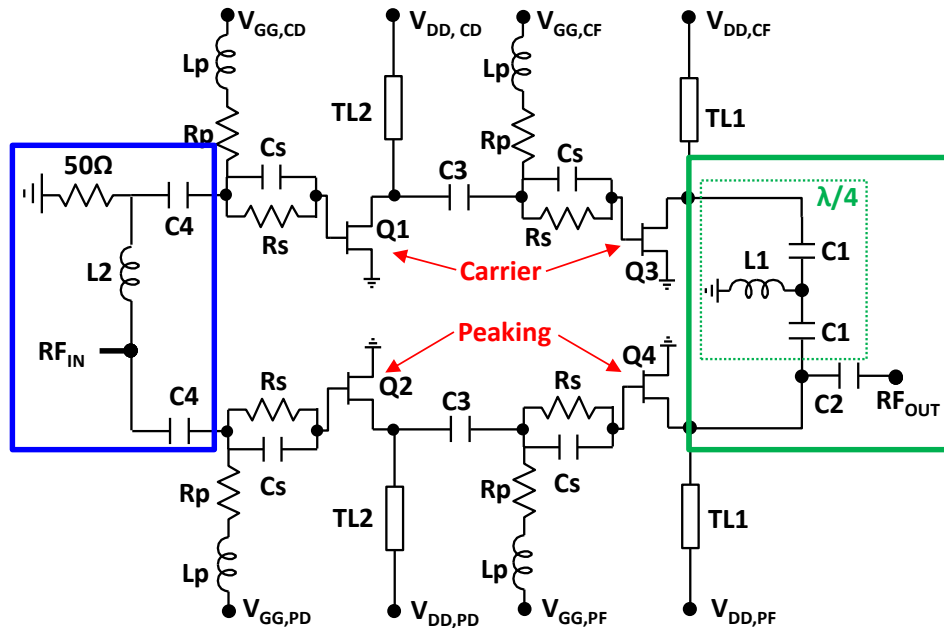
**Driver device:** same as in the final to avoid early saturation due to lower and lower  $G_{\text{out,CD}}$

**TL2-C3:** minimal InSMN to avoid any impedance inverting behaviour

**TL1 stub:** compensates the reactive part of  $Y_{\text{O,CF}}$  (i.e. it is  $B_L$ ) & brings drain bias.

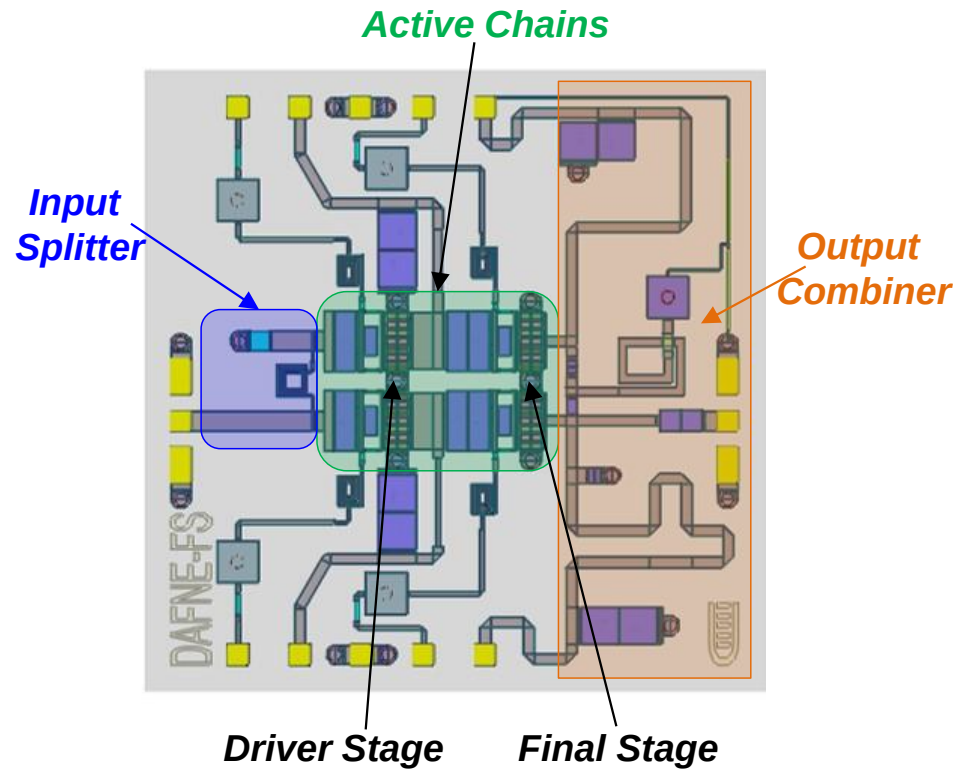


- The DPA architecture has been completed designing the output combiner and the input splitter.

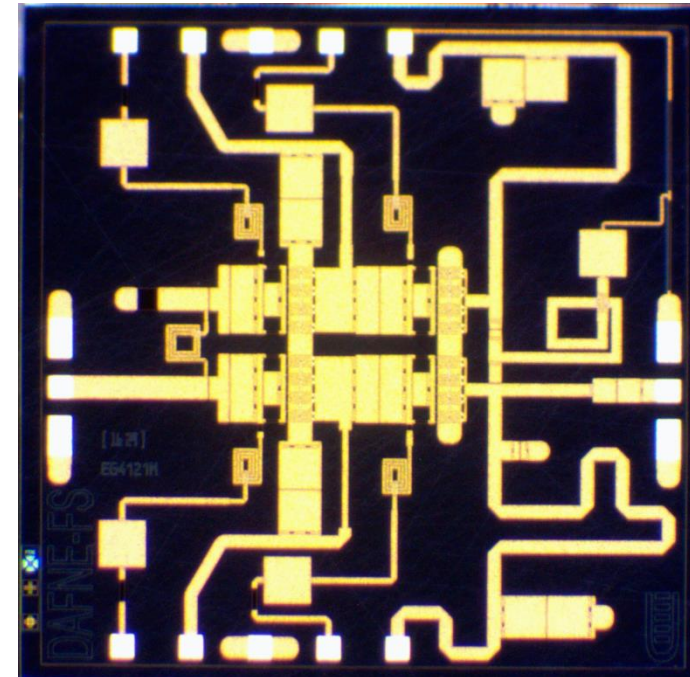


- Comparing the AM/PM of the final stage alone with the one achieved by the overall active chain, the AM/PM is reduced from 10° to just 3°

- The DPA has been designed in 0.25um GaN technology from Qorvo.
- Chip sizes are 3x3mm<sup>2</sup>



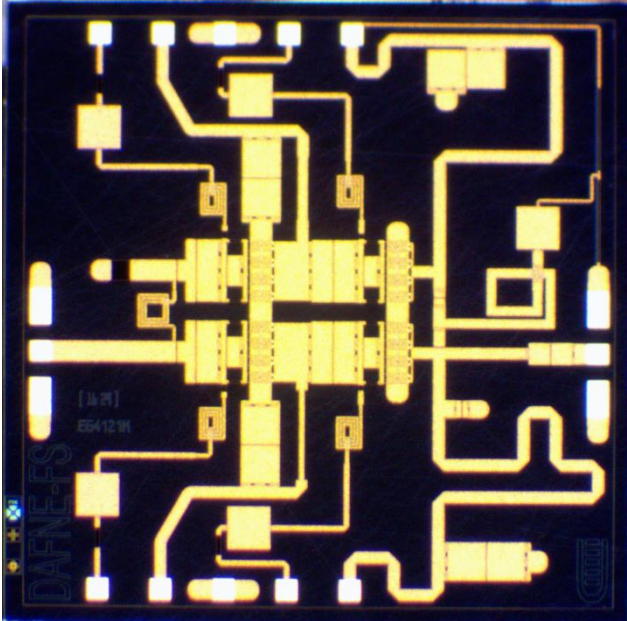
**MMIC Layout**



**MMIC photo**

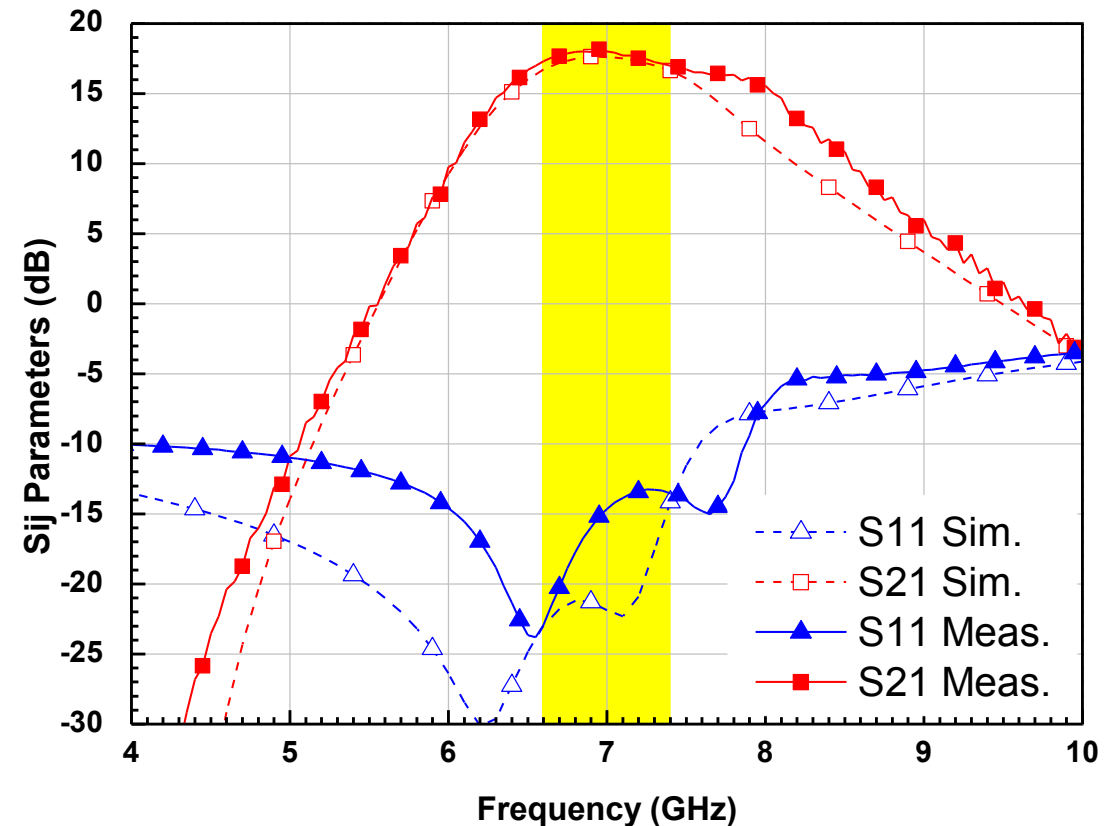
# Experimental Results

- Scattering parameters measurement results



## Bias Point

- $V_{DD,F}=28V$
- $V_{GG,CD}=-2.1V$
- $V_{DD,D}=7V$
- $V_{GG,PF}=-3V$
- $V_{GG,CF}=-2.6V$
- $V_{GG,PD}=-3.6V$

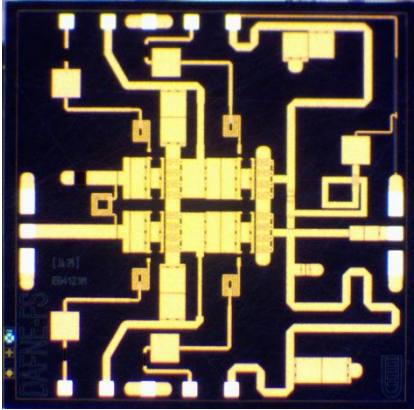


## From 6.7 to 7.3GHz

- $S_{11} < -13dB$
- $S_{21} > 16dB$

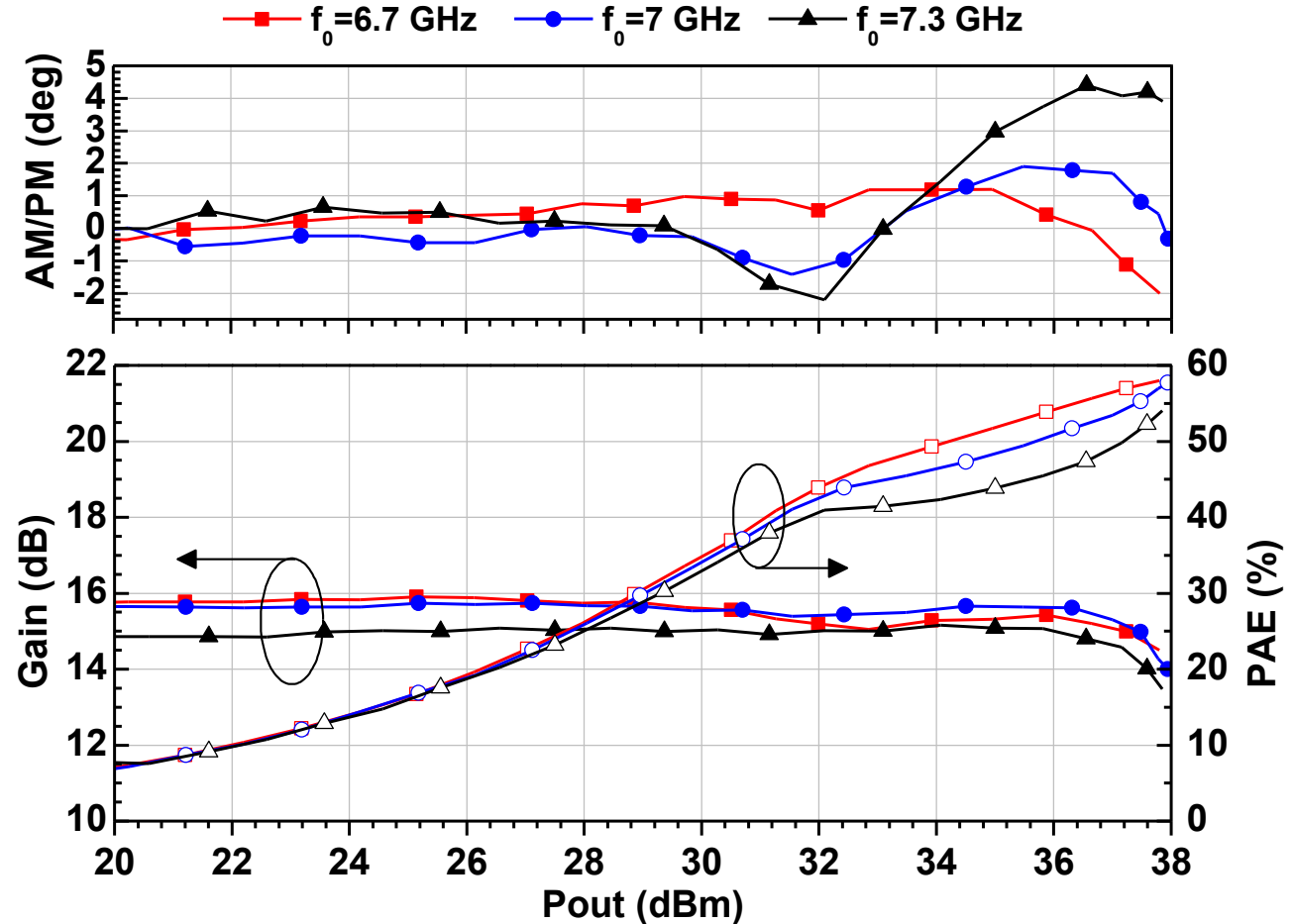
# Experimental Results

- Continuous wave measurement results.



*From 6.7 to 7.3GHz*

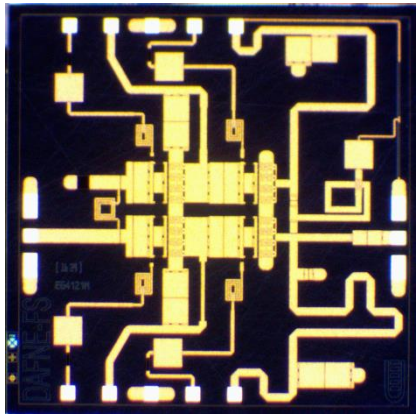
- $P_{out}=38\text{dBm}$
- $\text{Gain}>15\text{dB}$
- $\text{PAE}@7\text{dB}>40\%$
- $\text{PAE}@sat>56\%$
- $\text{AM/PM}<3^\circ$



- The proposed method is effective along a wider bandwidth too.

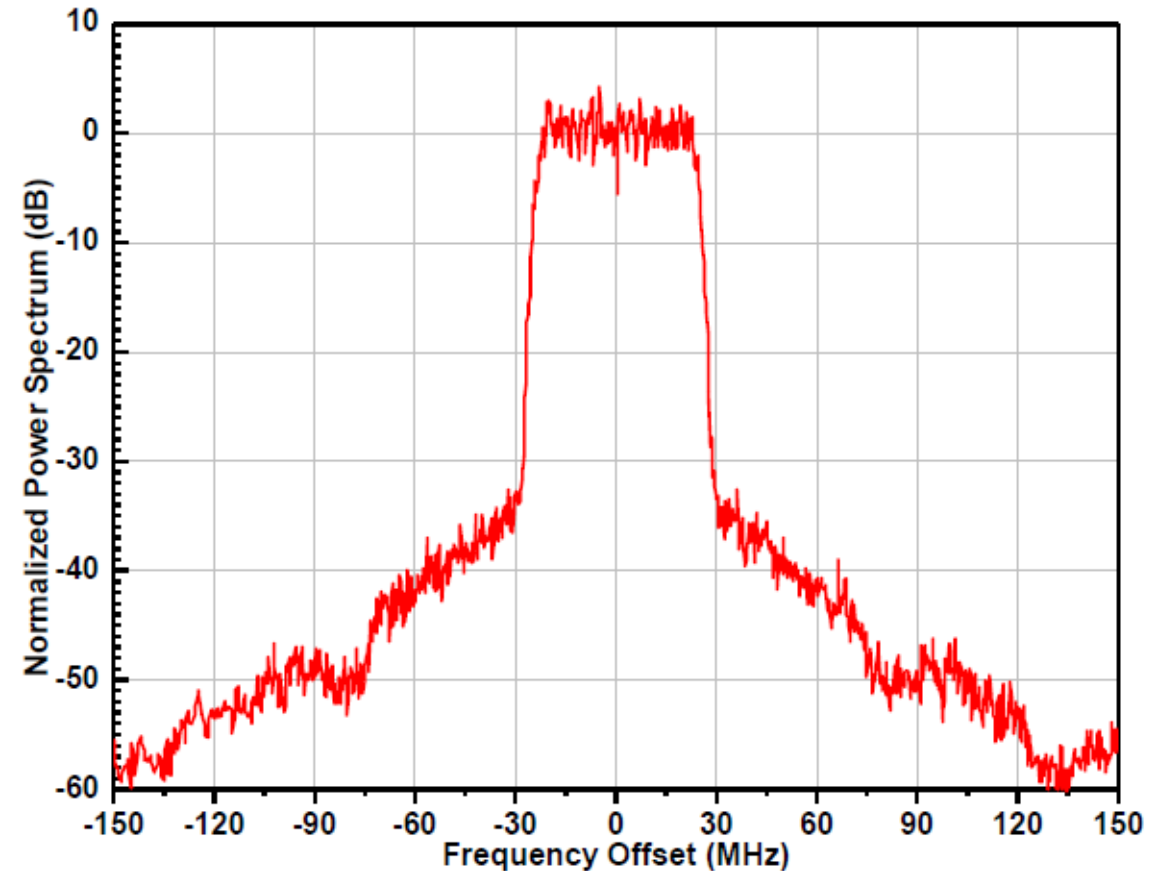
# Experimental Results

- The DPA has been tested with modulated signals (i.e., 4QAM modulation scheme and 7 dB of PAPR) having video bandwidths of 56 MHz.



**@7GHz & 32dBm**

- PAE > 40%
- $ACPR_{L/U} < -36$  dBc
- Without DPD



- Backhaul radio links, and more in general, microwave radio links are widely adopted in communication infrastructure, and they will be even more crucial in 5G network.
- **key Drivers will be:** Massive MIMO, larger bandwidth, higher efficiency and frequency of operation and better linearity.
- The DPA architecture can be efficiently adopted to maximize the efficiency versus linearity trade-off while assuring the fulfillment of the bandwidth requirement, even at millimeter waves.
- GaN technologies represent a suitable platform to develop microwave and millimeter waves electronics for next generation wireless communication systems.
- **Focus:** It has been demonstrated how the introduction of a non-linear driver stage in both Carrier and Peaking branches can be exploited to minimize the AM/PM of the DPA directly at chip level.

# GaN Doherty Amplifiers for Backhaul Radio Links

**Rocco Giofrè**

*E.E.Dept., University of Rome “Tor Vergata”, Rome, Italy*

[giofr@ing.uniroma2.it](mailto:giofr@ing.uniroma2.it)

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